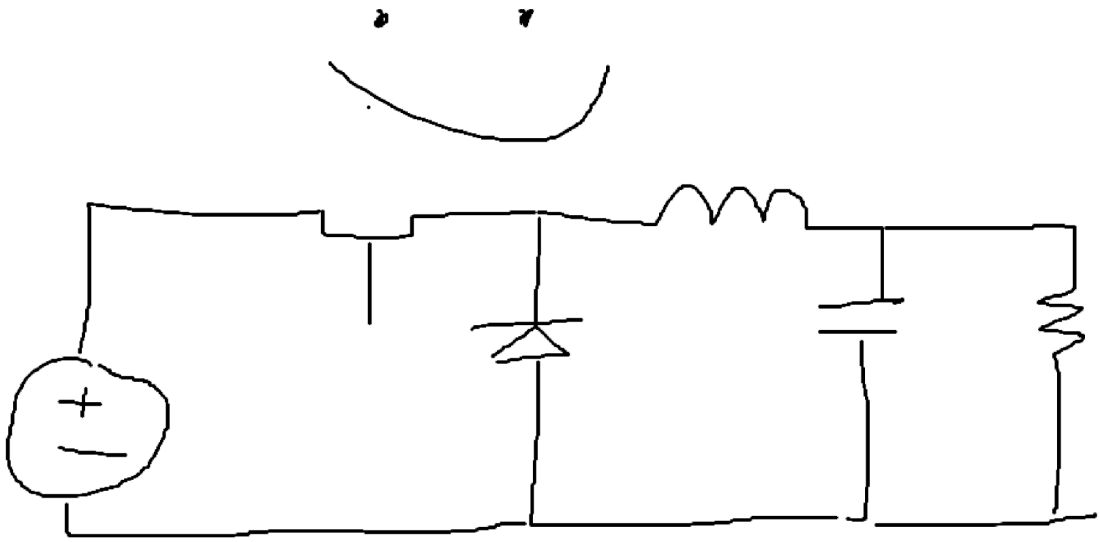
Formula Slug – Buck Converter Schematic Block
Power Electronics Onboarding Project 2026-2027

Overview

This project is centered around designing a buck converter to help power different boards on our LV system. This project requires you to design your own buck converter, sourcing components and creating the schematic and layout. You will also push it onto FS's github.



Our Low Voltage system consists of all our important logic and sensing boards, such as our Vehicle Control Unit, Telemetry boards, and Tractive Battery board. On all these boards, there are a variety of different components, who all may need to be powered by a different voltage level. For example, relays can operate at 12V, MCUs can operate at 5V, while some smaller chips run on as small as 3.3V. This is a big issue, because our low voltage bus runs on 12V!? If we overvolted those components, they would immediately stop working.

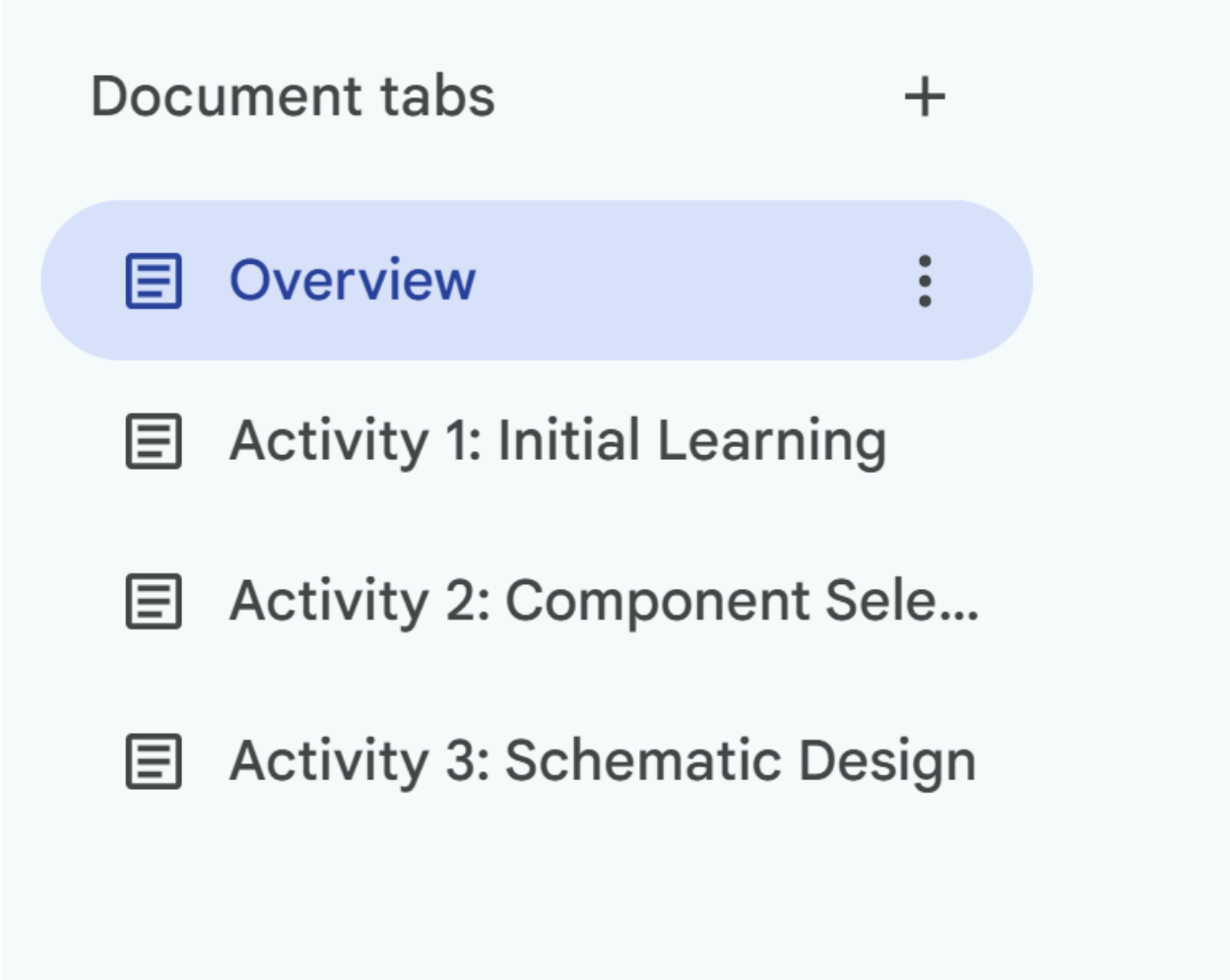
So what can we do to fix this?

A buck converter is used to transform a higher voltage to a lower voltage. We can use this to

power all of these components that require a much smaller power input, allowing it to properly function.

Navigating Project & Submission

There are 3 parts to this project: Initial Learning, Component Selection, and Schematic Layout. On the left side of the Google Doc, you will find the different tabs leading you to each activity.



Make a copy of this project, renaming it to "*Full Name* - Power Electronics Onboarding". Once every activity is completed, or before the due date, submit the document here. For example:

"Edmund Lai - Power Electronics Onboarding"

If there are any questions, message Edmund Lai on slack. There will be updates to this document if needed, check below for update logs.

Revision	Date	Updated Content
1.0	9/14/2026	Initial draft